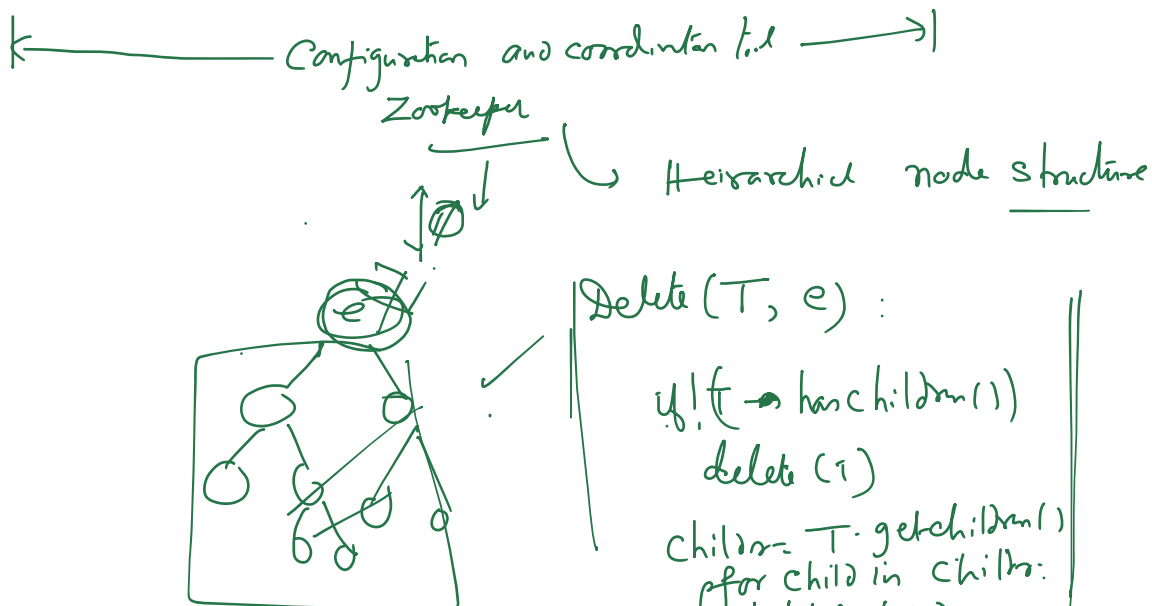
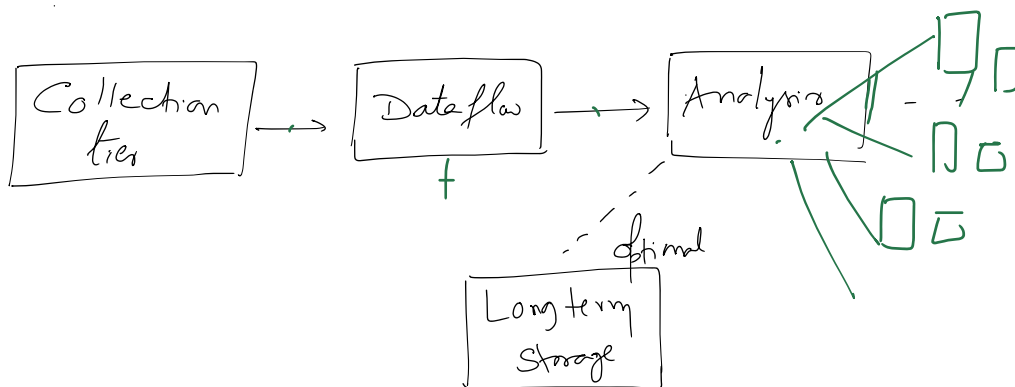




Delete(T, e):

if (e has children)

delete(e)

children = T.getchildren()

for child in children:

Delete(child)

exists(T, v):

if (match(T, v))

return T

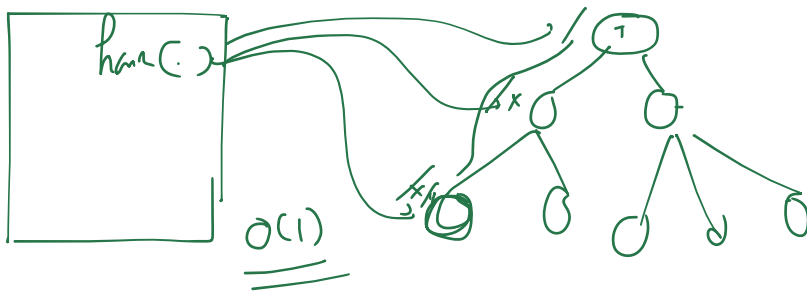
children = T.getchildren()

for child in children:

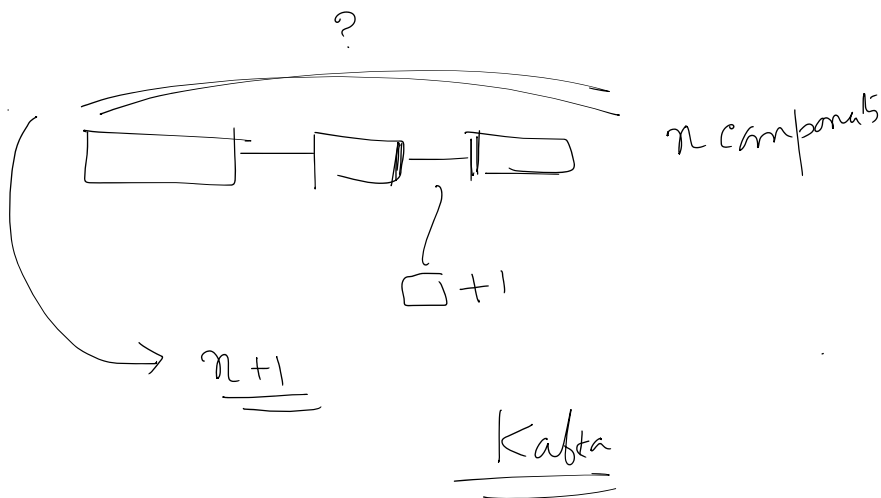
if (match(child, v))

return child

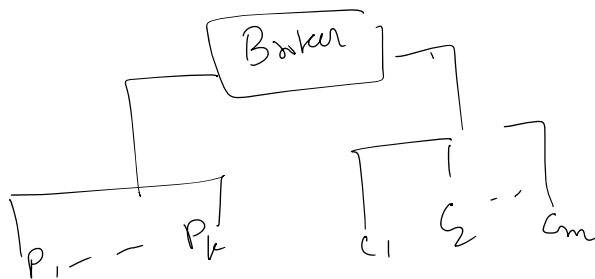
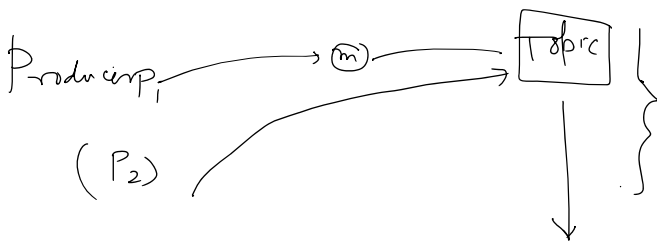
exists(child)



Non-Deterministic
probabilistic { freq
Cardinality
membership (check whether an event present in the stream)

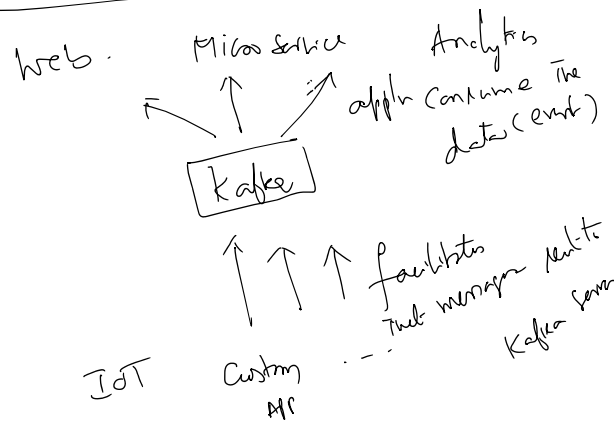


Topic: logical grouping of events



o Broker knows "topic"

Appn

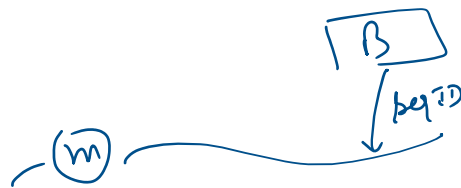
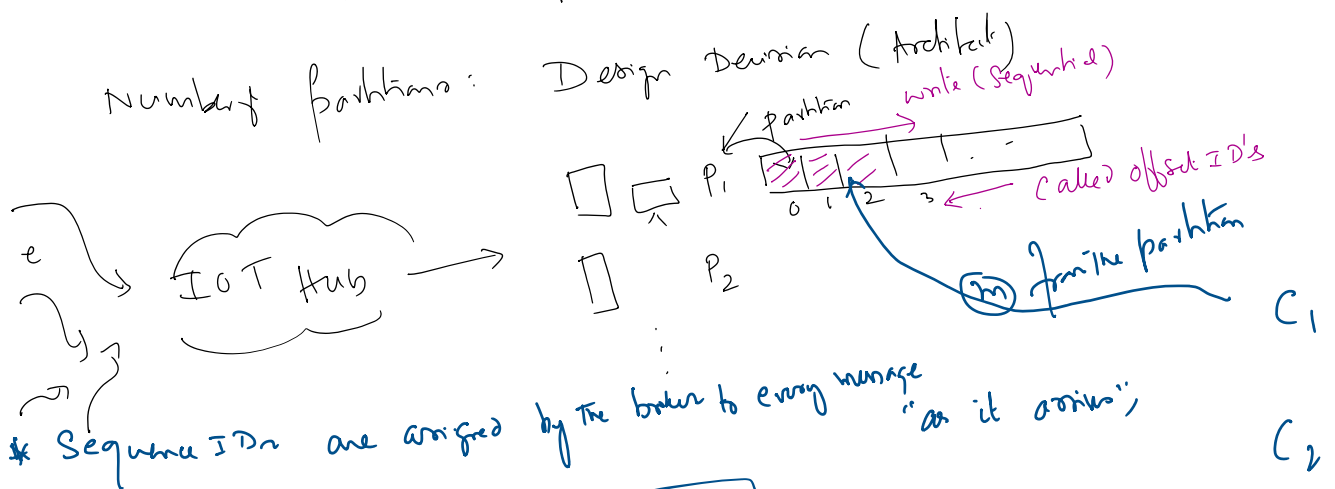


Communication happens through broker

Topic Partition:

Small independent partition of topic

Number of partitions: Design Decision (Architecture)



* messages are appended in a partition

locating a message?

topic name $\rightarrow$ partition number $\rightarrow$ offset id

Estimating number of partitions

$$N = \max \left\{ \frac{\text{Total System Throughput } (T_k)}{\text{Max. throughput of producer writing into single partition } (T_p)}, \frac{\text{Total System Throughput } (T_k)}{\text{Max. throughput of consumer reading the data from a single partition } (T_c)} \right\}$$

$$N = \max \left\{ \frac{T_k}{T_p}, \frac{T_k}{T_c} \right\}$$

In a streaming appn, the total throughput of streaming system is 42 GB/min

Throughput of producer is 5 MB/sec.

Estimate expected throughput consumer.

$$\begin{aligned} \text{Given } T_c &= 42 \text{ GB/min} = 42 \times 10^3 \text{ MB/min} \\ &= 42 \times 10^3 \text{ MB} / (60 \text{ sec}) \end{aligned}$$

$$\begin{aligned} \text{The throughput of producer} &= 5 \text{ MB/sec} \\ \Rightarrow T_p &= 5 \text{ MB/sec.} \end{aligned}$$

$$\begin{aligned} \frac{T_c}{T_p} &= \left(\frac{42 \times 10^3}{60} \right) / 5 = \frac{42 \times 10^3}{60 \times 5} = \frac{42 \times 10^3}{300} \\ &= \frac{4200}{3} = \boxed{1400} \end{aligned}$$

$$1400 = \max \left\{ 1400, \frac{T_c}{T_p} \right\}$$

$$T_c = \frac{42 \times 10^3}{60} = \frac{4200}{1}$$

$$\underline{1400} = \max \left\{ 1400, \frac{700}{2} \right\}$$

$$\frac{700}{2} \leq 1400$$

$$\frac{700}{140} \leq 2$$

$$2 > \frac{700}{2+400}$$

$$\boxed{x=5}$$

$$\geq \underline{5}$$

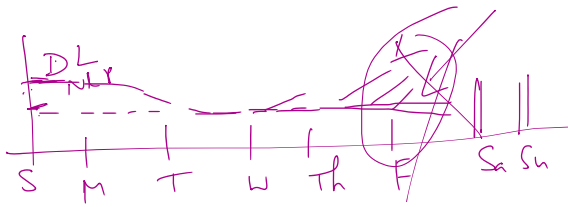
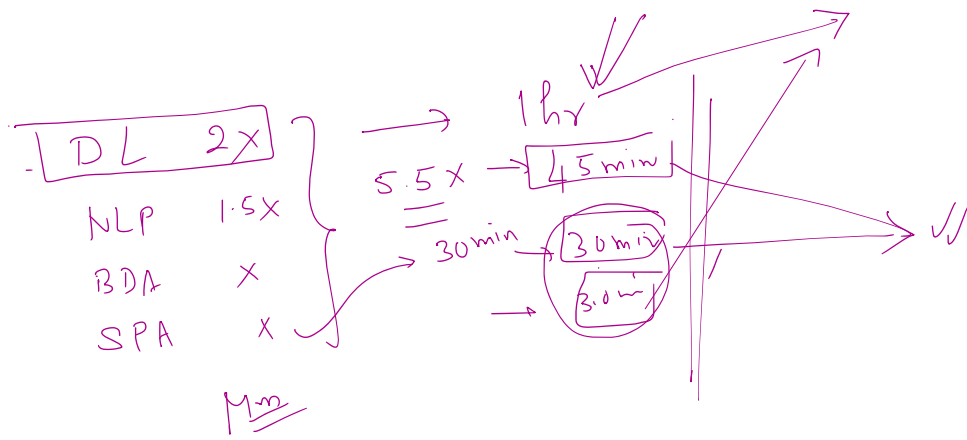
$$P_1 \rightarrow \boxed{m_1, m_2, m_3}$$

$$P_2: \boxed{m_4, m_5, x}$$

$$S_1 = \text{Set of elements}(P_1)$$

$$S_1 \cap S_2 = \emptyset$$

$$S_2 = \text{Set of elements}(P_2)$$



⊙ solve problems (NLP)
↳ before st

?